Discrete Random Variables

ANSWER KEY

Section 1: Probability Distribution Tables & Total Probability

1 The random variable X has probability distribution $P(X = x) = c(x^2 + 1)$ for $x = 0, 1, 2, 3$. Find the constant c.

a. $\frac{1}{18}$ ✓

b. $\frac{1}{14}$

c. $\frac{1}{15}$

d. $\frac{1}{20}$

For $x = 0 : 1c$; $x = 1 : 2c$; $x = 2 : 5c$; $x = 3 : 10c$. Sum $= (1 + 2 + 5 + 10)c = 18c = 1 \Rightarrow c = 1/18$.

2 A fair die is rolled. Let X be the number of rolls until a 6 appears. What kind of distribution does X follow?

a. **Geometric distribution** ✓

b. Binomial distribution

c. Uniform distribution

d. Poisson distribution

The number of independent Bernoulli trials until the first success is modeled by the Geometric distribution.

3 Given the probability distribution of Y: $P(Y = -1) = 0.3$, $P(Y = 0) = 0.4$, $P(Y = 2) = 0.3$. Find $P(Y \geq 0)$.

a. 0.7 ✓

b. 0.4

c. 0.3

d. 1.0

$P(Y \geq 0) = P(Y = 0) + P(Y = 2) = 0.4 + 0.3 = 0.7$.

Section 2: Expected Value $E(X)$ & Moments

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4 Find the expectation $E(X)$ for the distribution:
$P(X=1)=0.1, P(X=2)=0.3, P(X=3)=0.4, P(X=4)=0.2$.

a. 2.7 ✓

b. 2.5

c. 3.0

d. 2.8

$E(X)=1(0.1)+2(0.3)+3(0.4)+4(0.2)=0.1+0.6+1.2+0.8=2.7$.

5 If $E(X)=5$, what is $E(3X-4)$?

a. **11** ✓

b. 15

c. 19

d. 7

By linearity of expectation: $E(3X-4)=3E(X)-4=3(5)-4=15-4=11$.

6 A game charges 5 dollars to play. A fair coin is tossed: Heads pays 8 dollars, Tails pays nothing. What is the expected net gain for the player?

a. **-$1.00** ✓

b. +$1.00

c. -$2.00

d. +$4.00

Expected payout $=0.5(8)+0.5(0)=4$ dollars. Net gain $=4-5=-1.00$ dollar.

Section 3: Variance $\mathrm{Var}(X)$ & Standard Deviation

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7 Given $E(X) = 3$ and $E(X^2) = 13$, find $\text{Var}(X)$.

a. **4** ✓

b. 10

c. 2

d. 16

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 13 - 3^2 = 13 - 9 = 4$.

8 If $\text{Var}(X) = 6$, find the variance of $Y = 2X - 5$.

a. **24** ✓

b. 12

c. 7

d. 36

$\text{Var}(aX + b) = a^2\text{Var}(X)$. Here $\text{Var}(2X - 5) = 2^2 \times 6 = 4 \times 6 = 24$.

9 Calculate the standard deviation of X given $P(X = 0) = 0.5$ and $P(X = 2) = 0.5$.

a. **1** ✓

b. 2

c. 0.5

d. $\sqrt{2}$

$E(X) = 0(0.5) + 2(0.5) = 1$. $E(X^2) = 0(0.5) + 4(0.5) = 2$. $\text{Var}(X) = 2 - 1^2 = 1$. Standard deviation $\sigma = \sqrt{1} = 1$.

Section 4: Multi-Stage Discrete Problem Solving

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10 Two fair 4-sided dice (numbered 1 to 4) are rolled. Let S be the sum of the two numbers. Find $P(S > 5)$.

a. $\frac{6}{16} = \frac{3}{8}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{1}{4}$

Total outcomes $= 16$. Pairs with sum > 5: sum 6 has (2,4), (3,3), (4,2) (3 pairs); sum 7 has (3,4), (4,3) (2 pairs); sum 8 has (4,4) (1 pair). Total favorable $= 3 + 2 + 1 = 6$. Probability $= 6/16 = 3/8$.

11 For the sum S of the two 4-sided dice in the previous question, what is $E(S)$?

a. **5** ✓

b. 4

c. 6

d. 4.5

By symmetry, $E(S) = E(D_1) + E(D_2) = 2.5 + 2.5 = 5$.

12 A box has 3 red and 2 blue balls. Two balls are drawn without replacement. Let X be the number of blue balls drawn. Find $\text{Var}(X)$.

a. 0.36 ✓

b. 0.40

c. 0.48

d. 0.24

$P(X = 0) = \binom{3}{2}/\binom{5}{2} = 3/10 = 0.3$. $P(X = 1) = (3 \times 2)/10 = 6/10 = 0.6$.
$P(X = 2) = \binom{2}{2}/10 = 1/10 = 0.1$. $E(X) = 0(0.3) + 1(0.6) + 2(0.1) = 0.8$.
$E(X^2) = 0 + 1(0.6) + 4(0.1) = 1.0$. $\text{Var}(X) = 1.0 - 0.8^2 = 1.0 - 0.64 = 0.36$.

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